How Researchers Use AI

Day 1 - AI for Research: Common Uses and Limitations
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In this session

- The studies
- Adoption and frequency of use
- What researchers use AI for
- Who uses AI
- Benefits, concerns and regulation

Acknowledgement of Country

I would like to acknowledge the Traditional Owners of Australia and recognise their continuing connection to land, water and culture. The University of Sydney is located on the land of the Gadigal people of the Eora Nation. I pay my respects to their Elders, past and present.

The studies

Five recent studies

Study	Sample	When	Design
Van Noorden & Perkel (2023)	>1,600 researchers, worldwide	2023	Survey (self-selected sample)
Chugunova et al. (2026)	6,215 researchers, Max Planck and Fraunhofer, Germany	Jun 2024	Survey
Mishra et al. (2024)	226 clinical researchers, 59 countries (Harvard GCSRT)	Apr-Jun 2024	Survey
Mohammadi et al. (2026)	Publishing academics, 20 countries	Jan-Mar 2024	Survey
Khalifa & Albadawy (2024)	24 studies	2024	Systematic review

The authors of Van Noorden & Perkel (2023) and Mohammadi et al. (2026) note their samples are not representative of all researchers; Chugunova et al. (2026) report a sample close to their target populations.

Adoption and frequency of use

Awareness of AI tools

- Mishra et al. (2024): **87.6%** of clinical researchers were aware of large language models; those aware had a higher number of publications ($p < 0.001$).
- Mohammadi et al. (2026): **73.1%** of publishing academics had heard “a lot” about generative AI tools; only **3.4%** reported no familiarity.
- Chugunova et al. (2026): **42.4%** reported being very or rather familiar with AI tools; a further **44.0%** had used them a few times.

Frequency of use

- Chugunova et al. (2026): **25.9%** use AI tools at least once a day; **22.2%** never use them for any work task.
- Mohammadi et al. (2026): **54.7%** use generative AI at least monthly for academic purposes (**28.2%** never). For research specifically, **62.5%** use it at least monthly and **13.4%** never.
- Mishra et al. (2024): among aware respondents, **18.7%** had used LLMs in a publication; **81.3%** had not.
- Van Noorden & Perkel (2023): in 2023, none reported using AI regularly for research, and **17.4%** used generative AI weekly or more often.

> **Note**
>
> These are extraordinary figures for a technology that in practice didn't exist before November 2022.

Comparing the 2023 and 2024 surveys

Chugunova et al. (2026) report a shift from 2023 (Van Noorden & Perkel, 2023):

- In 2023, regular research use of AI was essentially absent; by mid-2024, a quarter of researchers used AI daily (Chugunova et al., 2026).

Reported non-use moved in the opposite direction:

> 14.6% of researchers in 2024 report not using AI tools, compared with 1.3% in 2023.

- Chugunova et al. (2026) attribute this to the 2023 sample being skewed toward researchers already interested in AI.

What researchers use AI for

Six domains of academic AI use

Khalifa & Albadawy (2024) group the literature into six domains:

Ideas and content

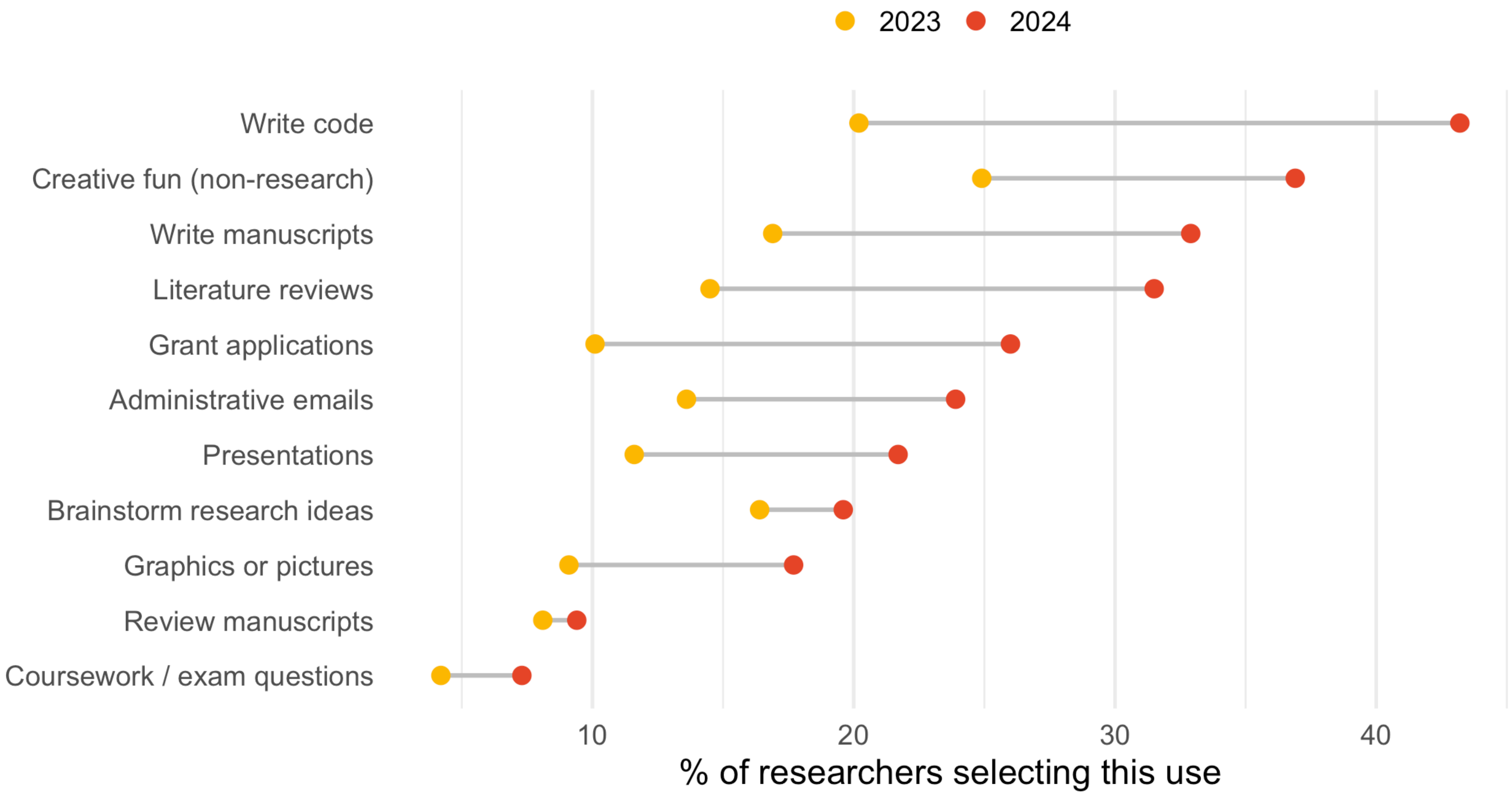
1. Idea development and research design
2. Content development and structuring
3. Literature review and synthesis

Data, output and ethics

4. Data management and analysis
5. Editing, review and publishing support
6. Communication, outreach and ethical compliance

Framework from Khalifa & Albadawy (2024).

Reported uses across the research process



Comparable use categories only. Source: Chugunova et al. (2026) and Van Noorden & Perkel (2023).

Most common research uses

- Chugunova et al. (2026): the two most frequent uses are **piloting and testing (47.9%)** and **writing code (43.2%)**; **50.6%** use AI for two to five distinct tasks.
- Mohammadi et al. (2026): the most common research uses are **translating text (13.5%)**, **proofreading drafts (13.0%)**, **synthesising initial drafts (12.5%)** and **literature reviews (12.5%)**; supporting data analysis is lower (**6.4%**). These tools were mainly used for academic writing rather than data analysis.
- Mishra et al. (2024): among those who had used LLMs, the most common stages were **grammatical errors and formatting (64.9%)**, **writing (45.9%)** and **revision and editing (45.9%)**.
- Van Noorden & Perkel (2023): in 2023, the single most common use was creative fun unrelated to research.

Disclosure of AI use

- Mishra et al. (2024): among researchers who had used LLMs in a publication, **40.5%** did not acknowledge that use.
- Mishra et al. (2024): **58.1%** thought journals should allow AI use in research, and **78.3%** thought regulations (for example modified journal policies, AI review boards, or detection tools) should be in place.

Who uses AI

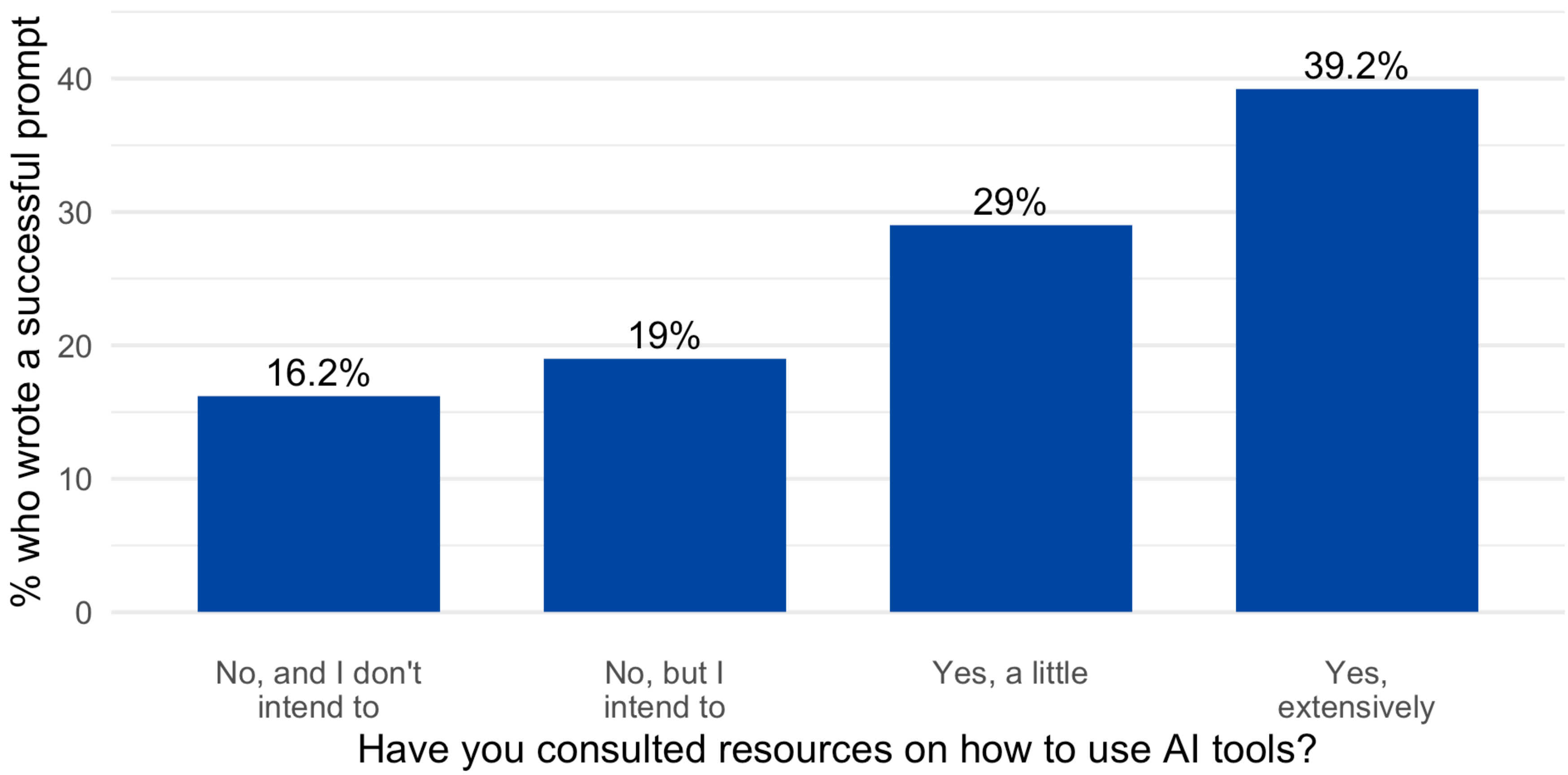
Role, field and age

- Chugunova et al. (2026): familiarity and use **decline with age** and **grow with education**; familiarity is **highest in the social sciences**. Clustering researchers as *leaders*, *builders* and *analysts*, “leaders” are the most positive toward AI.
- Mohammadi et al. (2026): **PhD students** are the highest users, followed by assistant professors and instructors. The **social sciences** show the highest adoption, then engineering. Highest adoption by country: **Taiwan, South Korea, India, Iran**; lowest: **United States, United Kingdom, China, Russia**.
- Mishra et al. (2024): only number of publications was significantly associated with awareness; age, region and field of practice were not.

The gender gap

- Chugunova et al. (2026): the gender gap in AI use for research is almost entirely explained by observable characteristics, with **familiarity the dominant factor (71% of the explained gap, 99% of the total gap)**. Among non-users, women rated AI as more helpful than men did. **45.9%** of women versus **34.8%** of men had not consulted, and did not intend to consult, resources such as prompt engineering guides.
- Mohammadi et al. (2026): women were **10% less likely** to use generative AI frequently (daily or weekly) for research, **4% less** for teaching, and **3% less** for administrative tasks.

Prompting as a skill



Chugunova et al. (2026): **50.4%** report using AI to speed up work, but only **21.0%** produced a successful prompt in a test task.

Benefits, concerns and regulation

Perceived future impact

- Van Noorden & Perkel (2023): more than half expected AI tools to become “very important” or “essential” in their field within a decade.
- Chugunova et al. (2026): **69.2%** expect AI to transform or revolutionise their field within a decade.
- Mishra et al. (2024): **52.0%** expect a major overall impact. Areas seen as most impacted: **grammatical errors and formatting (66.3%), revision and editing (57.2%)**, **writing (57.2%)** and **literature review (54.2%)**. Least impacted: methodology, journal selection and study ideas. **50.8%** anticipated a positive future impact; **32.6%** were unsure.

Perceived benefits

Mohammadi et al. (2026), share agreeing or strongly agreeing:

- Personalised tutoring or instruction: **37.4%**
- Improving problem-solving abilities: **35.2%**
- Enhancing students' learning: **34.5%**
- Fostering creativity and ideation: **31.2%** agree vs **35.9%** disagree
- Accuracy and reliability of AI content: **20.6%** confident vs **48.7%** disagree
- Consistency of AI content: **20.0%** agree vs **50.1%** disagree

Concerns about AI use

- Van Noorden & Perkel (2023) (machine learning): **69%** more reliance on pattern recognition without understanding; **58%** entrenching bias; **55%** fraud easier; **53%** irreproducible research. On generative AI: **68%** misinformation; **68%** easier plagiarism; **66%** mistakes in papers.
- Mohammadi et al. (2026) (moderate or extreme concern): inaccuracy **67.8%**; plagiarism **65.0%**; discouraging critical thinking **61.7%**; lack of transparency **59.2%**; lack of explainability **57.8%**; intellectual property **52.2%**; data privacy **49.0%**; bias **41.3%**.
- Mishra et al. (2024): misinformation **66.7%**; unintended bias **65.7%**; copyright infringement **63.1%**; impaired creativity **60.1%**; lack of accountability **58.1%**; data security **50.5%**.

Barriers and regulation

- Chugunova et al. (2026): the most common top-two barriers are **legal uncertainties (17.6%)**, **lack of knowledge (17.4%)** and **suitable tools not available (16.6%)**.
- Chugunova et al. (2026): on where guidance should come from, **58.7%** look to supranational bodies (for example the EU), **51.3%** to research societies, **49.0%** to professional associations.
- Mishra et al. (2024): **78.3%** thought regulations should be put in place; **58.1%** thought journals should allow AI use.

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